6.1 p.336 # 1, 5, 7, 13, 15, 17, 19, 23, 34

1. $u \cdot u = (-1)^2 + 2^2 = 5$ $\quad \frac{v \cdot u}{u \cdot u} = \frac{8}{5}$
$v \cdot u = 4(-1) + 6(2) = 8$

5. $\left(\frac{u \cdot v}{v \cdot v}\right) v = \left(\frac{-4+12}{4^2+6^2}\right)\begin{bmatrix}4\\6\end{bmatrix} = \frac{8}{52}\begin{bmatrix}4\\6\end{bmatrix} = \frac{2}{13}\begin{bmatrix}4\\6\end{bmatrix} = \begin{bmatrix}8/13\\12/13\end{bmatrix}$

7. $\|w\| = \sqrt{9+1+25} = \sqrt{35}$

13. $x - y = \begin{bmatrix}11\\2\end{bmatrix}$ $\quad \|x-y\| = \sqrt{121+4} = \sqrt{125} = 5\sqrt{5}$

15. $a \cdot b = -16 + 15 = -1$ not orthogonal

17. $u \cdot v = -12 + 2 + 10 + 0 = 0$ orthogonal

19. a) true . def. of $\|v\|$
b) true . . Thm 1C
c) true . Fig 5 disc.
d) false $(\text{Col}\,A)^\perp = \text{Nul}\,A^T$ counterex: $\begin{bmatrix}1&1\\0&0\end{bmatrix}$
e) true . box after ex 6

23. $u = \begin{bmatrix}2\\-5\\-1\end{bmatrix}$ $\quad v = \begin{bmatrix}-7\\-4\\6\end{bmatrix}$

$u \cdot v = -14 + 20 - 6 = 0$

$\|u\|^2 = u \cdot u = 4 + 25 + 1 = 30$

$\|v\|^2 = v \cdot v = 49 + 16 + 36 = 101$

$\|u+v\|^2 = (u+v) \cdot (u+v) = \begin{bmatrix}-5 & -9 & 5\end{bmatrix}\begin{bmatrix}-5\\-9\\5\end{bmatrix} = 25 + 81 + 25 = 131$

34. $A \sim \begin{bmatrix}1&0&5&0&-1/3\\0&1&1&0&-4/3\\0&0&0&1&1/3\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}$ $\quad R = \begin{bmatrix}1&0&5&0&-1/3\\0&1&1&0&-4/3\\0&0&0&1&1/3\end{bmatrix}$ $\quad N = \begin{bmatrix}-5&1\\-1&4\\1&0\\0&-1\\0&3\end{bmatrix}$

$x_1 = \frac{1}{3}x_5 - 5x_3$ $\quad x_2 = \frac{4}{3}x_5 - x_3$ $\quad x_4 = -\frac{1}{3}x_5$

6.2 p. 344 # 5*, 9*, 11*

5. $\begin{bmatrix}3\\-2\\1\\3\end{bmatrix}\begin{bmatrix}-1\\3\\-3\\4\end{bmatrix}\begin{bmatrix}3\\8\\7\\0\end{bmatrix}$ u_1 u_2 u_3

$u_1 \cdot u_2 = -3-6-3+12 = 0$

$u_1 \cdot u_3 = 9-16+7+0 = 0$

$u_2 \cdot u_3 = -3+24-21+0 = 0$

orthogonal set

9. $u_1 \cdot u_2 = -1+0+1 = 0$ $\therefore$ orthogonal set

$u_1 \cdot u_3 = 2+0-2 = 0$

$u_2 \cdot u_3 = -2+4-2 = 0$

nonzero so lin. indep by thm 4

3 vectors in $\mathbb{R}^3$ $\therefore$ basis for $\mathbb{R}^3$

$$x = \frac{x\cdot u_1}{u_1\cdot u_1}u_1 + \frac{x\cdot u_2}{u_2\cdot u_2}u_2 + \frac{x\cdot u_3}{u_3\cdot u_3}u_3$$

$$x = \frac{8+0-3}{1+0+1}u_1 + \frac{-8-16-3}{1+16+1}u_2 + \frac{16-4+6}{4+1+4}u_3$$

$$x = \frac{5}{2}u_1 - \frac{27}{18}u_2 + \frac{18}{9}u_3 = \frac{5}{2}u_1 - \frac{3}{2}u_2 + 2u_3$$

11. $y = \begin{bmatrix}1\\7\end{bmatrix}$, $u = \begin{bmatrix}-4\\2\end{bmatrix}$

$$\hat{y} = \frac{y\cdot u}{u\cdot u}\cdot u = \frac{-4+14}{16+4}\cdot u = \frac{10}{20}u = \frac{1}{2}\begin{bmatrix}-4\\2\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$$

6-3 p.352 #1*, 3*, 7*, 9, 11, 13, 15, 17, 19, 21

1. $\hat{x} = \frac{x\cdot u_1}{u_1\cdot u_1}u_1 + \frac{x\cdot u_2}{u_2\cdot u_2}u_2 + \frac{x\cdot u_3}{u_3\cdot u_3}u_3 = \frac{-16}{18}u_1 - \frac{8}{36}u_2 + \frac{12}{18}u_3$

$= -\frac{8}{9}\begin{bmatrix}0\\1\\-4\\-1\end{bmatrix} - \frac{2}{9}\begin{bmatrix}3\\5\\1\\1\end{bmatrix} + \frac{6}{9}\begin{bmatrix}1\\0\\1\\-4\end{bmatrix} = \frac{1}{9}\begin{bmatrix}0\\-18\\36\\-18\end{bmatrix} = \begin{bmatrix}0\\-2\\4\\-2\end{bmatrix}$

$z = x - \hat{x} = \begin{bmatrix}10\\-8\\2\\0\end{bmatrix} - \begin{bmatrix}0\\-2\\4\\-2\end{bmatrix} = \begin{bmatrix}10\\-6\\-2\\2\end{bmatrix} = 2u_4$ $\qquad x = \begin{bmatrix}0\\-2\\4\\-2\end{bmatrix} + \begin{bmatrix}10\\-6\\-2\\2\end{bmatrix}$

3. $u_1\cdot u_2 = -1+1+0$ ✓ orth. set $\qquad W = \text{span}\{u_1, u_2\}$

$\text{proj}_W y = \frac{y\cdot u_1}{u_1\cdot u_1}u_1 + \frac{y\cdot u_2}{u_2\cdot u_2}u_2 = \frac{3}{2}u_1 + \frac{5}{2}u_2 = \frac{1}{2}\left(\begin{bmatrix}3\\3\\0\end{bmatrix} + \begin{bmatrix}-5\\5\\0\end{bmatrix}\right)$

$= \begin{bmatrix}-1\\4\\0\end{bmatrix}$

7. $\text{proj}_W y = 0u_1 + \frac{28}{42}u_2 = \frac{2}{3}\begin{bmatrix}5\\1\\4\end{bmatrix} = \begin{bmatrix}10/3\\2/3\\8/3\end{bmatrix} = \hat{y}$

$z = y - \hat{y} = \begin{bmatrix}1\\3\\5\end{bmatrix} - \begin{bmatrix}10/3\\2/3\\8/3\end{bmatrix} = \begin{bmatrix}-7/3\\7/3\\7/3\end{bmatrix}$ $\qquad y = \hat{y} + z$

9. $\text{proj}_W y = \frac{6}{3}u_1 + \frac{10}{15}u_2 - \frac{2}{3}u_3 = \begin{bmatrix}2\\2\\0\\2\end{bmatrix} + \begin{bmatrix}-2/3\\2\\2/3\\-4/3\end{bmatrix} + \begin{bmatrix}2/3\\0\\-2/3\\-2/3\end{bmatrix} = \begin{bmatrix}2\\4\\0\\0\end{bmatrix} = \hat{y}$

$z = y - \hat{y} = \begin{bmatrix}4\\3\\3\\-1\end{bmatrix} - \begin{bmatrix}2\\4\\0\\0\end{bmatrix} = \begin{bmatrix}2\\-1\\3\\-1\end{bmatrix}$

11. $\text{proj}_W y = \frac{6}{12}v_1 + \frac{6}{4}v_2 = \frac{1}{2}\begin{bmatrix}3\\1\\-1\\1\end{bmatrix} + \frac{3}{2}\begin{bmatrix}-1\\1\\-1\\-1\end{bmatrix} = \frac{1}{2}\left(\begin{bmatrix}6\\-2\\2\\-2\end{bmatrix}\right) = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

$W = \text{span}\{v_1, v_2\}$, $v_1 \cdot v_2 = 0$ so orth.

13. $z \approx \text{proj}_W z = \frac{10}{15} v_1 + \frac{-7}{3} v_2 = \frac{1}{3}\left(\begin{bmatrix}4\\-2\\-6\\2\end{bmatrix} + \begin{bmatrix}-7\\-7\\0\\7\end{bmatrix}\right) = \begin{bmatrix}-1\\-3\\-2\\3\end{bmatrix}$

15. plane $= W = \text{span}\{u_1, u_2\}$

$\text{proj}_W y = \frac{35}{35} u_1 + \frac{-28}{14} u_2 = \begin{bmatrix}-3\\-5\\1\end{bmatrix} - \begin{bmatrix}-6\\4\\2\end{bmatrix} = \begin{bmatrix}3\\-9\\-1\end{bmatrix} = \hat{y}$

$z = y - \hat{y} = \begin{bmatrix}2\\0\\6\end{bmatrix}$ $\quad$ dist $= \|z\| = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$

17. a) $U^T U = \begin{bmatrix}2/3 & 1/3 & 2/3\\-2/3 & 2/3 & 1/3\end{bmatrix}\begin{bmatrix}2/3 & -2/3\\1/3 & 2/3\\2/3 & 1/3\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$ $\quad UU^T = \begin{bmatrix}8/9 & -2/9 & 2/9\\-2/9 & 5/9 & 4/9\\2/9 & 4/9 & 5/9\end{bmatrix}$

b) $U^T U = I_2$ so col. of U make orthonormal basis for W and by Thm 10

$\text{proj}_W y = UU^T y = \begin{bmatrix}8/9 & -2/9 & 2/9\\-2/9 & 5/9 & 4/9\\2/9 & 4/9 & 5/9\end{bmatrix}\begin{bmatrix}4\\8\\1\end{bmatrix} = \begin{bmatrix}2\\4\\5\end{bmatrix}$

19 $u_1 \cdot u_2 = 0$, $u_1 \cdot u_3 \neq 0$, $u_2 \cdot u_3 \neq 0$

proj u_3 on W, find orth. comp (v) which is $\perp$ W and $u_3 = v + \text{proj}$

$v = u_3 - \text{proj}_W u_3 = u_3 - \left(\frac{-2}{6} u_1 + \frac{2}{30} u_2\right) = \begin{bmatrix}0\\0\\1\end{bmatrix} + \begin{bmatrix}1/3\\1/3\\-2/3\end{bmatrix} + \begin{bmatrix}-1/3\\1/15\\-2/15\end{bmatrix} = \begin{bmatrix}0\\2/5\\1/5\end{bmatrix}$

21. a) True, u_1 and u_2 in W so $z \perp$ to them

b) True

c) False (proof thm 8)

d) True

e) True

6-4 p. 358 # 3*, 5, 7, 11, 13, 15, 17, 24

3. $u_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$ $u_2 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$ $v_1 = u_1$ $v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1$

$v_2 = u_2 - \frac{15}{30} v_1 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 3 \\ 3/2 \\ 3/2 \end{bmatrix}$

basis: $\begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix}$

5. $v_1 = u_1$ $u_1 = (1, -4, 0, 1)$ $u_2 = (7, -7, -4, 1)$

$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1 = u_2 - \frac{36}{18} v_1 = \begin{bmatrix} 7 \\ -7 \\ -4 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$

basis: $\begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$

7. $\|v_1\| = \sqrt{30}$ $\|v_2\| = \sqrt{54}$

orthonormal basis $= \left\{ \frac{1}{\sqrt{30}} v_1, \frac{1}{\sqrt{54}} v_2 \right\}$ $\frac{1}{3\sqrt{6}}$

$= \left\{ \begin{bmatrix} \sqrt{30}/15 \\ -\sqrt{30}/6 \\ \sqrt{30}/30 \end{bmatrix}, \begin{bmatrix} \sqrt{6}/3 \\ \sqrt{6}/6 \\ \sqrt{6}/6 \end{bmatrix} \right\}$

11. $u_1 = (1, -1, -1, 1, 1)$ $u_2 = (2, 1, 4, -4, 2)$ $u_3 = (5, -4, -3, 7, 1)$

$v_1 = u_1$

$v_2 = u_2 - \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1 = u_2 + \frac{5}{5} v_1 = u_2 + v_1 = (3, 0, 3, -3, 3)$

$v_3 = u_3 - \frac{u_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{u_3 \cdot v_2}{v_2 \cdot v_2} v_2 = u_3 - \frac{20}{5} v_1 + \frac{12}{36} v_2 = u_3 - 4v_1 + \frac{1}{3} v_2$

$= \begin{bmatrix} 5 \\ -4 \\ -3 \\ 7 \\ 1 \end{bmatrix} + \begin{bmatrix} -4 \\ 4 \\ 4 \\ -4 \\ -4 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}$

basis: $\{v_1, v_2, v_3\}$

13. $R = Q^T A = \begin{bmatrix} 5/6 & 1/6 & -3/6 & 1/6 \\ -1/6 & 5/6 & 1/6 & 3/6 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}$

15. $v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$ $\|v_1\| = \sqrt{5}$ $\quad v_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}$ $\|v_2\| = 6$ $\quad v_3 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}$ $\|v_3\| = 4$

$$Q = \begin{bmatrix} 1/\sqrt{5} & 1/2 & 1/2 \\ -1/\sqrt{5} & 0 & 0 \\ -1/\sqrt{5} & 1/2 & 1/2 \\ 1/\sqrt{5} & -1/2 & 1/2 \\ 1/\sqrt{5} & 1/2 & -1/2 \end{bmatrix}$$

$$Q^T A = \begin{bmatrix} 1/\sqrt{5} & -1/\sqrt{5} & -1/\sqrt{5} & 1/\sqrt{5} & 1/\sqrt{5} \\ 1/2 & 0 & 1/2 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix} = R$$

17. a) False, ex. 2 but scale nonzero b) True, #1 of Thm 11 c) True, ex 4

24. $\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}$

$v_1 = x_1$

$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = x_2 + \frac{400}{400} v_1 = x_2 + v_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}$

$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 = x_3 + \frac{200}{400} v_1 + \frac{48}{36} v_2 = x_3 + \frac{1}{2} v_1 + \frac{4}{3} v_2$

$= \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}$

$v_4 = x_4 - \frac{x_4 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_4 \cdot v_2}{v_2 \cdot v_2} v_2 - \frac{x_4 \cdot v_3}{v_3 \cdot v_3} v_3 = x_4 - \frac{200}{400} v_1 + \frac{36}{36} v_2 + \frac{54}{108} v_3$

$= x_4 - \frac{1}{2} v_1 + v_2 + \frac{1}{2} v_3$

$= \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}$

basis: $\{v_1, v_2, v_3, v_4\}$

6.5 p. 366 #3, 5, 7, 9, 15

3. $A^TA = \begin{bmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}$

$A^Tb = \begin{bmatrix} 1 & -1 & 0 & 2 \\ -2 & 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$

$[A^TA \mid A^Tb] = \begin{bmatrix} 6 & 6 & 6 \\ 6 & 42 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 1 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 6 & -2 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4/3 \\ 0 & 1 & -1/3 \end{bmatrix}$ $\quad \hat{x} = \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix}$

5. $A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$

$A^Tb = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}$

$\begin{bmatrix} 4 & 2 & 2 & 14 \\ 2 & 2 & 0 & 4 \\ 2 & 0 & 2 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 1 & 1 & 7 \\ 1 & 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & -1 & 1 & 3 \\ 0 & -1 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\quad x_1 = 5 - x_3$ $\quad x_2 = -3 + x_3$ $\quad \hat{x} = \begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

7. $A\hat{x} = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix}$ $\quad b - A\hat{x} = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$

$\|b - A\hat{x}\| = \sqrt{1^2 + 3^2 + (-3)^2 + 1^2} = \sqrt{20} = 2\sqrt{5}$

9. $b = \begin{bmatrix} 4 \\ -2 \\ -3 \end{bmatrix}$ $a_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ $a_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$

a) $\hat{b} = \frac{b \cdot a_1}{a_1 \cdot a_1} a_1 + \frac{b \cdot a_2}{a_2 \cdot a_2} a_2 = \frac{4}{14} a_1 + \frac{6}{42} a_2$

$= \frac{2}{7} a_1 + \frac{1}{7} a_2 = \frac{1}{7}\left(\begin{bmatrix} 2 \\ 6 \\ -4 \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}\right) = \frac{1}{7}\begin{bmatrix} 7 \\ 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

b) $A\hat{x} = \hat{b}$

$$\begin{bmatrix} 1 & 5 & 1 \\ 3 & 1 & 1 \\ -2 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 1 \\ 0 & -14 & -2 \\ 0 & 14 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 1 \\ 0 & 1 & 1/7 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2/7 \\ 0 & 1 & 1/7 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 2/7 \\ 1/7 \end{bmatrix}$$

15. $\hat{x} = R^{-1}Q^T b$ or $Rx = Q^T b$

$$\hat{x} = \frac{1}{3}\begin{bmatrix} 1 & -5 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 2/3 & 2/3 & 1/3 \\ -1/3 & 2/3 & -2/3 \end{bmatrix}\begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 7/3 & -8/3 & 11/3 \\ -1 & 2 & -2 \end{bmatrix}\begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 12 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

6.6 p.374 #1,3,7,9

1. (0,1) (1,1) (2,2) (3,2)

$$X=\begin{bmatrix}1&0\\1&1\\1&2\\1&3\end{bmatrix} \quad Y=\begin{bmatrix}1\\1\\2\\2\end{bmatrix} \quad X^TX=\begin{bmatrix}1&1&1&1\\0&1&2&3\end{bmatrix}\begin{bmatrix}1&0\\1&1\\1&2\\1&3\end{bmatrix}=\begin{bmatrix}4&6\\6&14\end{bmatrix}$$

$$X^TY=\begin{bmatrix}1&1&1&1\\0&1&2&3\end{bmatrix}\begin{bmatrix}1\\1\\2\\2\end{bmatrix}=\begin{bmatrix}6\\11\end{bmatrix}$$

$$\begin{bmatrix}4&6&6\\6&14&11\end{bmatrix}\sim\begin{bmatrix}2&3&3\\0&5&2\end{bmatrix}\sim\begin{bmatrix}2&3&3\\0&1&\frac{2}{5}\end{bmatrix}\sim\begin{bmatrix}2&0&\frac{9}{5}\\0&1&\frac{2}{5}\end{bmatrix}$$

$$\sim\begin{bmatrix}1&0&9/10\\0&1&2/5\end{bmatrix} \qquad y=\frac{9}{10}+\frac{2}{5}x$$

3. (-1,0) (0,1) (1,2) (2,4)

$$X=\begin{bmatrix}1&-1\\1&0\\1&1\\1&2\end{bmatrix} \quad Y=\begin{bmatrix}0\\1\\2\\4\end{bmatrix} \quad X^TX=\begin{bmatrix}1&1&1&1\\-1&0&1&2\end{bmatrix}\begin{bmatrix}1&-1\\1&0\\1&1\\1&2\end{bmatrix}=\begin{bmatrix}4&2\\2&6\end{bmatrix}$$

$$X^TY=\begin{bmatrix}1&1&1&1\\-1&0&1&2\end{bmatrix}\begin{bmatrix}0\\1\\2\\4\end{bmatrix}=\begin{bmatrix}7\\10\end{bmatrix}$$

$$\begin{bmatrix}4&2&7\\2&6&10\end{bmatrix}\sim\begin{bmatrix}1&3&5\\4&2&7\end{bmatrix}\sim\begin{bmatrix}1&3&5\\0&-10&-13\end{bmatrix}\sim\begin{bmatrix}1&3&5\\0&1&13/10\end{bmatrix}$$

$$\sim\begin{bmatrix}1&0&11/10\\0&1&13/10\end{bmatrix} \qquad y=\frac{11}{10}+\frac{13}{10}x$$

a)

7. $X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}$ $Y = \begin{bmatrix} 1.8 \\ 2.7 \\ 3.4 \\ 3.8 \\ 3.9 \end{bmatrix}$ $\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$

b) $X^TX = \begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix}$ $X^TY = \begin{bmatrix} 52.1 \\ 201.5 \end{bmatrix}$

$$\begin{bmatrix} 55 & 225 & 52.1 \\ 225 & 979 & 201.5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.760 \\ 0 & 1 & -.199 \end{bmatrix}$$

$y = 1.76x - .20x^2$

9. $(1, 7.9)$ $(2, 5.4)$ $(3, -.9)$

$y = A\cos x + B\sin x$

$X = \begin{bmatrix} \cos 1 & \sin 1 \\ \cos 2 & \sin 2 \\ \cos 3 & \sin 3 \end{bmatrix}$ $Y = \begin{bmatrix} 7.9 \\ 5.4 \\ -.9 \end{bmatrix}$

$X^TX = \begin{bmatrix} 1.445 & -.063 \\ -.063 & 1.554 \end{bmatrix}$ $X^TY = \begin{bmatrix} 2.912 \\ 11.431 \end{bmatrix}$

$$\begin{bmatrix} 1.445 & -.063 & 2.912 \\ -.063 & 1.554 & 11.431 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2.342 \\ 0 & 1 & 7.447 \end{bmatrix}$$

$y = 2.342\cos x + 7.447\sin x$

6.7 p. 382 #1,3,5,7,9,13,16,21,23

1. $\langle u,v\rangle = 4u_1v_1 + 5u_2v_2$ $\quad x=(1,1)$ $\quad y=(5,-1)$

a) $\|x\| = \sqrt{4(1)(1)+5(1)(1)} = \sqrt{9} = 3$

$\|y\| = \sqrt{4(5)(5)+5(-1)(-1)} = \sqrt{100+5} = \sqrt{105}$

$|\langle x,y\rangle|^2 = (4(1)(5)+5(1)(-1))^2 = (20-5)^2 = 15^2 = 225$

b) $4(5)z_1 + 5(-1)z_2 = 0 \rightarrow 20z_1 = 5z_2 \rightarrow 4z_1 = z_2$

multiples of $\begin{bmatrix}1\\4\end{bmatrix}$

3, 5, 7: P_2; $t = -1, 0, 1$

3. $p(t) = 4+t$, $q(t) = 5-4t^2$

$\langle p,q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$

$= 3(1) + 4(5) + 5(1)$

$= 28$

5. $\|p\| = \sqrt{3^2+4^2+5^2} = \sqrt{50} = 5\sqrt{2}$

$\|q\| = \sqrt{1^2+.5^2+1^2} = \sqrt{27} = 3\sqrt{3}$

7. $\text{proj}_p q = \dfrac{\langle p,q\rangle}{\langle p,p\rangle} p = \dfrac{28}{50} p = \dfrac{14}{25}(4+t) = \dfrac{56}{25} + \dfrac{14}{25}t$

9. P_3, $t = -3, -1, 1, 3$ $\quad p_0(t)=1$, $p_1(t)=t$, $p_2(t)=t^2$

$p_0 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$ $p_1 = \begin{bmatrix}-3\\-1\\1\\3\end{bmatrix}$ $p_2 = \begin{bmatrix}9\\1\\1\\9\end{bmatrix}$

a) $\text{proj}_{p_0,p_1} p_2 = \dfrac{\langle p_2,p_0\rangle}{\langle p_0,p_0\rangle} p_0 + \dfrac{\langle p_2,p_1\rangle}{\langle p_1,p_1\rangle} p_1$

$\hat{p}_2 = 5$ $\quad = \dfrac{9+1+1+9}{1+1+1+1} p_0 + \dfrac{-27-1+1+27}{9+1+1+9} p_1 = 5p_0 + 0p_1 = 5p_0$

b) $q = p_2 - \hat{p}_2 = t^2 - 5$ orth. to p_0 and p_1 and $\{p_0, p_1, q\}$ orth. basis for span$\{p_0, p_1, p_2\}$. At $(-3,-1,1,3)$, q is $(4,-4,-4,4)$. So scale by $\frac{1}{4}$ and $q = \frac{1}{4}(t^2-5)$

16. If $\{u, v\}$ is an orthonormal set in V, then $\|u-v\| = \sqrt{2}$

$$\begin{aligned}\|u-v\|^2 &= \langle u-v, u-v\rangle = \langle u, u-v\rangle - \langle v, u-v\rangle \\ &= \langle u,u\rangle - \langle u,v\rangle - \langle v,u\rangle + \langle v,v\rangle \\ &= \langle u,u\rangle - 2\langle u,v\rangle + \langle v,v\rangle \\ &= \|u\|^2 - 2\langle u,v\rangle + \|v\|^2\end{aligned}$$

since $\{u,v\}$ orthonormal, $\|u\|^2 = \|v\|^2 = 1$ and $\langle u,v\rangle = 0$
so $\|u-v\|^2 = 2$ and $\|u-v\| = \sqrt{2}$

21. $$\int_0^1 (1-3t^2)(t-t^3)\,dt = \int_0^1 (t - t^3 - 3t^3 + 3t^5)\,dt$$

$$= \int_0^1 (t - 4t^3 + 3t^5)\,dt = \left(\tfrac{1}{2}t^2 - t^4 + \tfrac{1}{2}t^6\right)\Big|_0^1 = \tfrac{1}{2} - 1 + \tfrac{1}{2} = 0$$

23. $$\sqrt{\int_0^1 (1-3t^2)^2\,dt} = \sqrt{\int_0^1 (1-6t^2+9t^4)\,dt}$$

$$= \sqrt{\left(t - 2t^3 + \tfrac{9}{5}t^5\right)\Big|_0^1}$$

$$= \sqrt{\left(1 - 2 + \tfrac{9}{5}\right)} = \sqrt{4/5} = 2/\sqrt{5}$$